GENETIC PARADOX IN OBESITY: EXPLORING THE APOB/APOA1 RATIO IN CARDIOVASCULAR RISK ASSESSMENT

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INTRODUCTION

Obesity is a known cardiovascular risk factor, yet some patients with severe obesity present with normal lipid profiles. Standard lipid panels may overlook residual risk. The ApoB/ApoA1 ratio is a stronger predictor of cardiovascular events, as shown in the AMORIS and INTERHEART studies. This ratio may reveal hidden risk in seemingly low-risk patients. We present a 39-year-old male with class 3 obesity (BMI 90) and a normal lipid panel. His elevated ApoB/ApoA1 ratio highlighted potential cardiovascular risk, underscoring the need for improved risk assessment in severe obesity.

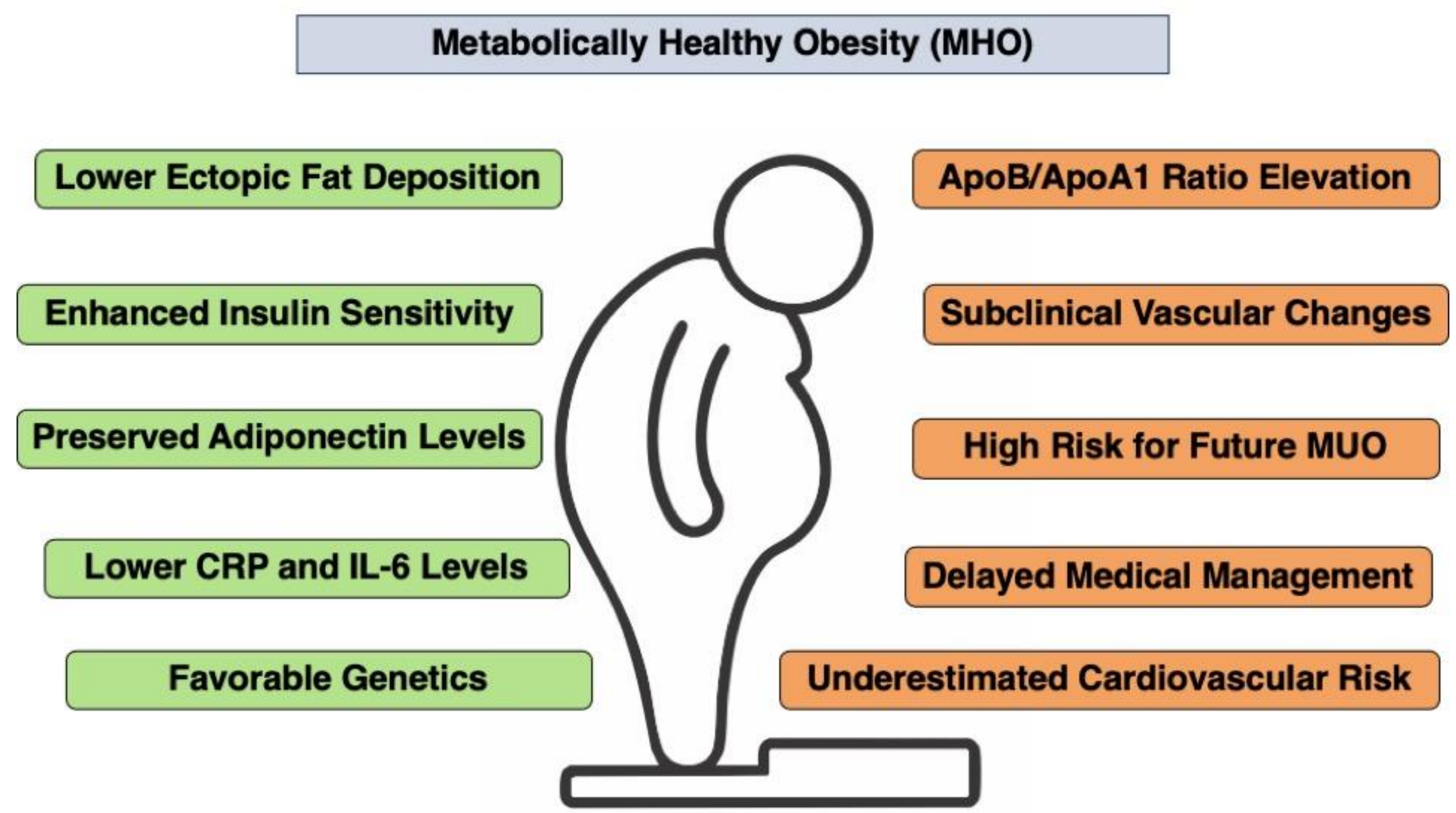
CASE PRESENTATION

We present the case of a 39-year-old male with class 3 obesity (BMI 90.60, weight 227.20 kg) who presented with a stage III gluteal ulcer that had been draining blood and pus for two weeks. He had no prior medical history, no home medications, and no family history of adverse cardiac outcomes. His social history was unremarkable, and he denied tobacco, alcohol, or illicit drug use. Despite his extreme obesity, initial lab work revealed a normal lipid panel, which remained normal on repeat testing. Apolipoprotein testing showed an ApoB/ApoA1 ratio of 0.72, indicating moderate cardiovascular risk.

Parameter	Value	Reference Range
BMI	**90.6**	< 25 (normal)
Weight	**227.2 kg**	60.2 kg – 81.1 kg
Total Cholesterol	**77 mg/dL**	< 200 mg/dL
Triglycerides	**60 mg/dL**	< 150 mg/dL
HDL	**22.4 mg/dL**	> 40 mg/dL (men)
LDL	**43 mg/dL**	< 100 mg/dL
ApoB	**52 mg/dL**	< 90 mg/dL (optimal)
ApoA1	**72 mg/dL**	> 120 mg/dL
ApoB/ApoA1 Ratio	**0.72**	0.64 - 0.80 (Moderate Risk)

DISCUSSION

- A subset of individuals with obesity exhibit a Metabolically Healthy Obesity (MHO)– normal blood lipids, blood pressure, and glycemic control despite excessive adiposity.
- Multiple landmark studies, including the AMORIS study, INTERHEART study, Québec Cardiovascular Study, and Copenhagen General Population Study, have demonstrated that the ApoB/ApoA1 ratio can predict risk even in those with normal lipid profiles.
- Based on thresholds established in these studies, our patient's ApoB/ApoA1 ratio (0.72) places him in a moderate-risk category, reflecting an increased burden of atherogenic particles despite normal LDL-C.
- Favorable genetic traits such as IRS1 variants, familial hypobetalipoproteinemia, PCSK9 loss-of-function mutations, ANGPTL3 mutations, CETP deficiency, ABCA1 mutations, LPIN1 variants, and APOA5 variants may explain these findings.



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DISCLOSURES

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